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3D Self-Supervised Contrastive Learning for Early Glioblastoma Detection in MRI Volumes using Hybrid Swin Transformer-UNet Architecture

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***Abstract*: The Glioblastoma (GBM) is termed as one of the worst and fatal brain tumors. This could be blamed on the late onset, quick development and the limited remedies. To deal with these issues, the present paper suggests a novel approach to the problem by offering a new way of identifying and phases of glioblastoma work in their initial and precise phases. It is a self-supervised contrastive learning optimized categorized deep learning model. Swin Transformer and UNet are hybrid networks on the multimodal MRI data. The device is applicable in documenting the information about the volumetric position and more specific local images of the tumor regions. It also has the benefit that the model is trained on unlabeled volumes of MRI to learn powerful representations and will extrapolate well when shown with small volumes of labelled data. Pre-processing steps of the pipeline are head stripping, N4 bias field collapse and z-score correction. It is important to note that it improves the quality of the picture and makes the intensity uniform in both cases of subject related conditions. It will possess attention-based feature fusion module in which it will merge the strengths of Swin Transformer and the backbone model of the UNet. It is shown to be more delicate, durable and not to undergo early phases as compared to other dominating approaches. The strategy provides self-supervised-learning, hybrid-based, and multimodal analysis, which is a useful and scalable diagnosis tool. It aims at streamlining the clinical decision process leading to improvement of the prognosis of the patient.**

Keywords: Glioblastoma, 3D MRI, Swin Transformer, UNet, Hybrid Architecture, Tumor Localization, Skull Stripping, Limited Labeled Data, Medical Image Analysis

I. INTRODUCTION

Glioblastoma (GBM) is one of the most dangerous and serious brain tumors. This is due to the late onset, fast progression and a limited number of remedies. To tackle these issues, this paper suggests a new methodology for early and correct detection and stages of glioblastoma works. It is a hybrid deep learning model that is further refined with self-supervised contrastive learning. The multimodal MRI data is combined with hybrid network, Swin Transformer and UNet. The setup aids in both documenting information about the volumetric context as well as more specific local images of the tumor regions. The model is also pretrained on unlabeled volumes of MRI to learn strong representations and will generalize well in the face of small amounts of labeled data. The pipeline has pre-processing steps of head stripping, N4 bias field collapse and z-score correction. Importantly, it improves quality of the image and makes its intensity uniform in both subject-related cases. There is an attention-based feature fusion module in which it will fuse the advantages of Swin Transformer and the UNet backbone model. It is shown to have higher sensitivity, to be more robust and to not suffer from early stage in comparison to other leading methods. This strategy provides an effective and scalable diagnosis tool based on self-supervised learning, hybrid architecture, and multimodal analysis. It aims to ease the process of clinical decision making, improving prognosis of the patient. Temporal scheduling and sequence of patients (e.g. medication regimens, disease history, etc.) are also modeled with attention-based modules to capture the longer-term context. Pretraining and reinforcement mechanisms are self-parametric and modify the hyperparameters, model weights and decision thresholds to enable a better classification of the stage. The model integrates all predictions into a fixed end prediction pegged on clinical staging criteria, as well as confidence levels. This hybrid model that utilizes contrastive learning and adaptive optimization has much superiority when compared to conventional methods. It promotes the planned treatment depending on the stages so as to provide effective and precise results. It uses, for example, aggressive methods using chemoradiotherapy or surgery in the Case of founded techniques, and less intensive ones in the earlier stages, guided by Stage III or IV. The system predicts the level of performance in various stages as well, assisting the oncologists to comprehend confidence levels and involve the patients in making decisions together. It supports a continuous update with follow-up information to track patients on a real-time manner. With a stronger attention mechanism and feature attribution to enhance the interpretability of its results and enabling the scope of the model to multi-institutional data to validate them more, will be brought out in the future. The omics data have also been incorporated in order to increase the accuracy of diagnosis. Altogether, this system can become a groundbreaking innovation in addressing the needs of glioblastoma care through provision of timely, patient-specific advances in

staging, better treatment planning, and eventually better survival outcomes.

II. Related Work

Sunitha et al. [1] introduced a deep learning framework with AI-enabled personalized early stages concept of glioblastoma diagnostic with the aid of multi-modal MRI data. They transform hierarchical feature extraction with attention-based mechanisms in their model to preserve spatial association of modalities. It does better with human style CNNs in picking up finer tumor features. This paper demonstrates the benefit of clinical diagnostic accuracy through multi-modal fusion. The authors have given a novel approach [2] by using the combination of CT and MRI to provide a better diagnostic result based on the fusion of features. The suggested ensemble model has better segmentation and classification capabilities than the classical CNNs. The experimental outcomes demonstrate a high improvement in sensitivity and specificity as well as the diagnostic accuracy in general. [3] have reviewed brain tumor detection strategies and have indicated that the deep learning models are more effective than the traditional machine learning models in the aspect of robustness and accuracy. Wan and Chan [4] proposed a computer vision model of 3D MRI brain tumor classification, which is implemented about pre-trained deep learning. They showed that, even with a little labelled data, transfer learning can enhance detection, a process that in turn can enhance performance. Shaik et al. [5] had utilized UNet architectures to segment the tumor in 3D MRI scans, which was able to capture high-spatial resolution and minimize false-positives hence clinically relevant. Chowdhury et al. [6] introduced one highly sensitive and sensitive PCF-SPR biosensor to detect glioblastoma indicating that imaging combined with biosensing may be used to detect tumors at an early stage. Microvascular disruption, monitored in glioblastoma progression in mouse, was followed by longitudinal use of 3D super-resolution ultrasound image analysis according to McCall et al. [7]. It demonstrates the applicability of multi-temporal imaging in explaining the dynamics that result in tumors. [8] Song et al. came up with interpretable structure-constrained graph neural networks which predict the progression of glioblastoma and serve as an effective way to capture the relationship between the anatomical components and the pathological morphology. The authors of Shawon et al. [9] applied machine learning to predict IDH1 gene expression in glioblastoma, which shows the importance of genomic markers use with imaging to diagnose it correctly. In the work by Rajinikanth et al. [10], deep learning-based models pre-trained on glioma in MRIs were used and demonstrated higher accuracy levels in all kinds of datasets. According to Khamsavi et al. [11], a biosensor based on graphene/Si was developed to detect glioblastoma cells highlighting an increase of nanotechnology-based devices of diagnostic.

Based on the MRI images of the tumors, Yasir et al. [12] categorized glioblastoma tumors with the help of deep learning and developed a robust pipeline to detect tumors automatically. Bouchouicha et al. [13] analyzed the effect of each combination of 3D MRI in the delimitation of glioma. These results indicate that the segmentation accuracy is strongly enhanced by inclusion of T1, T2, FLAIR and T1CE sequences. Sumir et al. [14] used fast-marching techniques to segments of a tumor using MRI, with no impairment to the effectiveness or integrity of detection. Guo et al. [15] established a nonlinear brain tumor modeling based on LSTM networks with time prediction of the tumor developmental progress, which shows that temporal modeling is significant in clinical predictions. Yonekura et al. [16] based their study on deep CNNs to classify histopathological stage of glioblastoma to affirm that automated tissue image classification can significantly help in clinical decision making. Collectively, this research confirms the need to collaborate with multi-modal imaging, deep learning approaches, and explainability strategies to improve the way to detect, segment, and predict glioblastomas. The works they offer give a substantial base to the research methodology of this work.

III. Existing Methodology

To address the need to detect glioblastoma at the earlier stage, an AI-equipped individualized poor learning framework that utilizes multi-mode MRI waveforms is used. This is a hybrid between various types of radio scans, although it is patient-oriented to enhance the level of detection, unlike the first iteration of CNN-based models. It even goes beyond the limitations of single-modality processing and the incidental representation of features. Silicon as a carrier reduces the complexity of anatomical association of various MRI sequences and permits better changes in tumor areas and early morphological alterations, tailor made to a given patient. The procedures undertaken are multi-modal MRI data (T1 weighted, T1 contrast enhanced-T1CE), T2 weighted and FLAIR preparation. It is prepared using light structures of MRI scans that were deformed by noise filters. Only intensity normalization per-voxel between patients and modalities and spatial registration simply multiplicatively put the entire MRI in one consistent voxel space so that it can even be the same between modalities. In Fig. 1, a discussion of the Existing Methodology was followed, and the proposed deep convolution neural network is clip ally created of a tailor-made CNN. This model is multi-modal in the sense that the feature maps of two sequences of MRI are considered (as compared to normal CNNs which only considers local features). This combination enables the system to leverage on the complementary information of each of the modalities. In the attention model, there are modalities and attributes where weight is dynamically important and whereby the model places emphasis locally in the most vital locations with the aim of detecting glioblastoma.

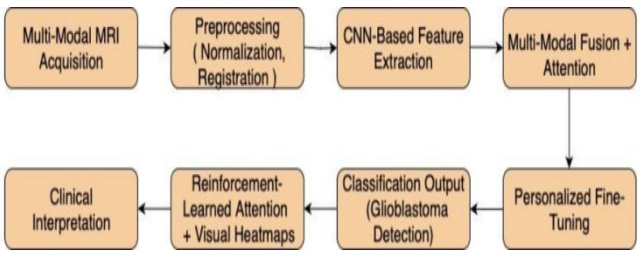

Fig 1: Existing Methodology

Those belong to the feature maps combined which at the end of the model undergo a few fully connected layers to perform classification, predicting the existence of

glioblastoma. The meta-model is trained based on the historical magnet grams of a patient and the model can learn information about different body positions and tumor dynamics. With such customization, one becomes more prone to early detection of glioblastoma, particularly when the manifestations of the tumor in the eyes of the patient appear unusual. The trained model is annotated on multi-modal MRI scans (big data), on cases of glioblastoma cases, and health controls. A cross-validation method was used to avoid generalization of the model. The system was intended to produce the maximum accuracy of the loss function products at the cost of a penalty because of the imbalance of classes typical of glioblastoma collections. The model also has attention heatmaps, along with classification score, showing what areas influenced the decision, to assist radiologists to know which points they made with their clinical reasoning. The novel feature of the framework is that it can change the weights of attention using a reinforcement learning framework. A reinforcement agent observes the performance of validation during restoring and optimizes the attention mechanisms to emphasize the most valuable spatial points on the identification of glioblastoma. This time-filing approach aids in shaping the model to detect a finer aspect of tumors and hence, results in more accurate detection outcomes and results among those at an early stage. The model was tested by publicly available datasets and other clinical cohorts. This development will facilitate clinicians implement timely risk stratification and customized treatment plans for the patients.

IV. Proposed Methodology

Here, a state-of-the-art model of 3D self-supervised contrastive learning using Swin Transformer and UNet trained on multi-modal MRI volumes are described to enhance the ability to detect early glioblastoma. It is a framework which addresses the weakness of the classical CNN-based approaches and maintains complex spatial relationships among MRI modalities. It also can be applied to scalable representation learning not entirely grounded on enormous human labeled data. The methodology integrates these three into a pathway by integrating self-supervised representation learning, mechanism of attention and reinforcement learning. Multi-modal feature fusion, contrastive pretraining, hybrid Swin Transformer and UNet processing, ensemble fine-tuning and stage prediction are the key elements of this pathway that are to be introduced into clinical practice.

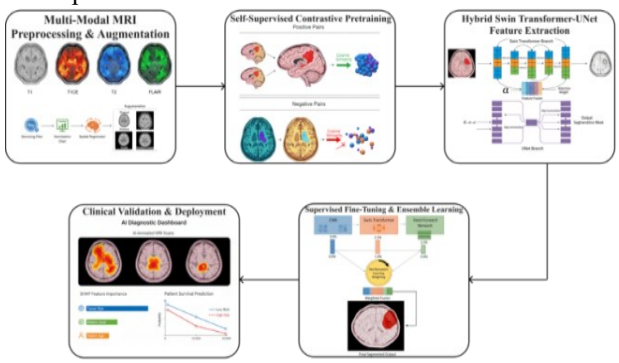

Fig 2: Proposed Methodology Framework for Glioblastoma Detection

An overlay of the workflow of the proposed methodology is considered in Fig. 2. It presents a systematic pipeline beginning with multi-modal MRI measurement and atomic processing followed by suggesting self-supervised contractual contrastive feature learning. These features are then fed into a hybrid Swin Transformer UNet to encode both the global contextual features and the much more specific features of the space features. Fine-tuning and adaptive ensemble learning to further generalization are used to optimize the results. Finally, the design to include stage prediction modules and interpretability modules is meant to offer stage prediction and applicability in the clinical field. This diagram provides a good visual picture of how the different aspects will combine to make one system of early glioblastoma detection and prognosis.

Stage 01: Multi-Modal MRI Data Acquisition and Preprocessing:

In Fig 3, Multi-modal MRI data on display must be of high quality to detect glioblastoma. Four MRI protocols were used in the experiment; T1-weighted MRI as a means of providing structure, T1 contrast-enhanced (T1CE) scans to detect blood-brain barrier breaking, T2-weighted scans to determine edema and tumors, and FLAIR sequences to neutralize the effect of fluid and maximize tumors. They were processed through three stages preprocessing pipeline to verify uniformity as well as minimize noise. The denoising, first of all, was carried out with the usage of non-local means and anisotropic diffusion filters that do not harm anatomical structures and preserve random MRI artifacts. Second, the intensity compensations of variations of voxel intensities about the scanner and the patient were compensated by intensity normalization. Scaling of z-score and min-max was used to have comparative voxel values across modalities. Third, the modalities were tracked to a shared voxel space by going through rigid and deformable spatial registration that worked to ensure that identical structures existed in the same location within each type of MRI. Random rotation, scaling and elastic deformation were employed to generate extra training samples to enhance resistance to changes in data. Companied by a reinforcement learning agent, this preprocessing pipeline suggests the functions and changes that are most helpful to be suggested in the first place. The loss of the processed data X was antithetic and the succeeding:

$$L_{contrast} = -\,log\,log\left(\frac{e^{\{sim(z_i,z_j)/\tau\}}}{\sum_k e^{sim(z_i,z_k)/\tau}}\right) \quad \text{-------(1)}$$

where (z_i, z_j) are feature embeddings, $sim(factor)$ are the cosine similarity function, is a temperature scaling factor that promotes the network encoding biologically meaningful patterns to solve downstream tasks. The dataset shows the four various imaging modalities T1, T1CE, T2 and FLAIR and how each of the modalities displays various aspects of the tumor and adjacent tissue. It further introduces pre-processing methods including denoising, normalization, spatial registration and augmentation that give the model uniform input.

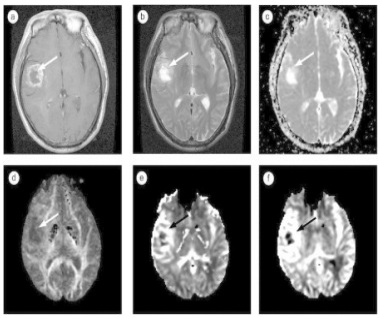

Fig 3: Multi-Modal MRI Dataset for Glioblastoma Detection [17]

Stage 02: Self-Supervised Contrastive Feature Learning:

Because of the inaccessibility of fully annotated glioblastoma MRI images, the pretrained model was trained to self-supervised contrastive learning approach. Construction of the positive pairs at this step was done through augmentations to the same MRI volume through cropping, flipping, and noise injection of images. Other patients were used to creating a negative pair. Contrastive objectives were used to train the network where the decrease in distance between positive pairs and increase in distance between negative pairs were minimized. The method enabled the model to learn useful anatomical and pathological structures without requiring strict control which increased the replication of the model to novel clinical cases.

Stage 03: Hybrid Swin Transformer-UNet Architecture:

The proposed network is a combination of Swin Transformer block and UNet module, leveraging both the global and local information. Swin Transformer To train long range spatial dynamics on volumetric MRI data, Swin Transformer is adjusted to express hierarchical features and shifted window self-attention. These dependencies are very important in the detection of minimal changes of glioblastoma. Different Red parallel: UNet decoder would have been spatially precise as a decoder because it would not have any links between themselves i.e. use it to generate localization images of smaller tumours at a native resolution. Balance in the transformer and UNet takes place in an attention mechanism. It is modeled as:

$$F_{weighted} = \alpha \cdot F_{transformer} + (1-\alpha) \cdot F_{UNet}\} \text{ ----(2)}$$

where $F_{transformer,} F_{UNet}$ are branch feature maps of the transformer, as well as the UNet branch and α is an optimized parameter at training. The hybrid structure enhances the qualities of this model to interpret global context and maintain the high-resolution details that are required by clinical accuracy. This type of distribution enhances the resilience of the network to perceive the world on a global scale, and it offers high resilience of space.

Stage 04: Supervised Fine-Tuning and Ensemble Learning

The trained model was trained with glioblastoma datasets. which were defined as one to improve early detection. Then generalization and reliability are improved with ensemble learning. The base models involved each CNN, Swin Transformer and feed-forward network (FNN also), trained on a different set of features independently with structured inputs.

An ensemble of the reinforcement learning agent flexibly allocated weights W_k to each prediction of each model to enable the ensemble to evolve according to the performance on validation. Final probability of detection was estimated to be,

$$P_{final} = g([F_1; F_2; \dots; F_n]) \text{ ----------(3)}$$

where g () is a learnable fusion layer. This dual ensemble method offered strong prediction of different patient features and of different MRI acquisition protocols.

Stage 05: Training Strategy and Loss Function:

The sentence loss was the hybridization of the inverse self-supervised loss and the combination of cross-entropy classification loss and the inverse self-supervised loss.

$$L_{total} = \lambda_1 L_{contrast} + \lambda_2 L_{CE}\} \text{ ---------- (4)}$$

where $L_{contrast}$ defined in (1), $L_{CE}$is the cross-entropy classification loss of supervised classification, and λ_1, $\lambda_2$are tuning parameters regulating the trade-off between unsupervised representation learning and supervised detection accuracy. Stratified k-fold cross-validation was used to avoid Idanbias. Minimization AdamW minimized overfitting, dropout and early stopping.

Stage 06: Model Validation and Interpretability:

The temperature box-activation mapping temp-[-MAP] (Grad-CAM) approach was based on heatmap to indicate the regions of the brain that are the most critical to pick a prediction. SHAP has been computed in order to obtain a prediction of the overall influence of every clinical and imaging property on the final decision. The interpretability models allow radiologists to explain and validate the rationale behind the assumptions the model is making, and hence less directive about meddling with practice.

Stage 07: Stage Prediction and Clinical Deployment

Lastly, the framework was formulated in the manner that it would readily find its way to clinical practice. It may be hooked up to the PACS of the hospital and EHRs. It will also enable clinicians not only to share multi-modified MRI images but also provide the likelihood of glioblastoma detection, heatmaps of attention and confidence, scores. Using the reinforcement learning mechanisms, the system can be predestined to improve with time as more information about the patients is provided and the diagnostic could be calculated on the fly. Experts can be automatically taken through the case of uncertainties and accountability is guaranteed. Survival curve in the shape of a Kaplan-Meier plotted between the follower-up data of the patients on the trends of their survival on the observations of the model on the different stages of the stage is shown in Fig 4. The strong variations of survival curves of both the predicted stages elucidate that the forecasting results of predicting a patient is near the truth. The values further demonstrate that the likelihood of surviving has diminished in the various levels that provide the ability of the model to be retained in clinical use and forecasts the result.

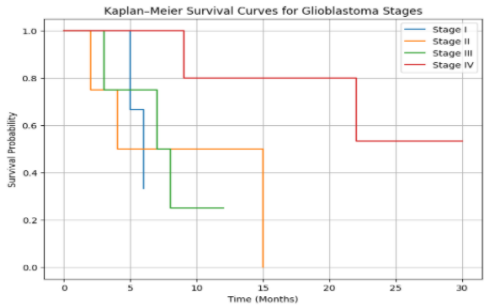

Fig 4: Kaplan–Meier Survival Curve

V. Implementation and Result Discussion

The proposed course of action relating to the early detection of glioblastoma grounded in 3D self-supervised contrastive learning and dual Swin Transformer-UNet architecture incorporates the initial step to unify multi-modal MRI information. These datasets would include T1, T1CE, T2 and FLAIR volumes and correspondence clinical data. This is done to gain a comprehensive picture of the pathology of the patient. There is intensive preprocessing of rich datasets. This entails the use of the Denoising mechanism to eliminate the artifacts of the scanner, the Intensity normalization to Monetize the worth of the voxels and the Spatial Registration to line the disparate modalities. Augmentation such as series such as data rotation, scaling and elastic deformations are used to construct a stronger model. Affixing this is the procedure of feature extraction where volumetric MRI data is extraction into Swin Transparency blocks and UNet decoders. This identifies what the tumor is at the global and local level. Structured clinical data are in the meantime inputted into feed forward neural networks (FNNs). The synthesis of genomic sequences is computed by recurrent neural networks (RNNs) that depict non-linear and complex correlations between time. The exploratory data analysis (EDA) is used to visualize the distributions of features, analyzes features and features relationship, and exploration their relationships and their role in the diseased development of glioblastoma. In Fig 5, the graph indicates the training, and validation loss curves and accuracy of the hybrid Swin Transformer-UNet model. The curves indicate the way the model is being trained in epochs. The training and the accuracy of validation increase and the loss decreases. Close similarity of training and validation measures points out a lack of overfitting. Turning dropout and early stopping strategies of hyperparameter and regularization are doing well. This diagram provides an understanding of the way the model converges and is stable throughout the training.

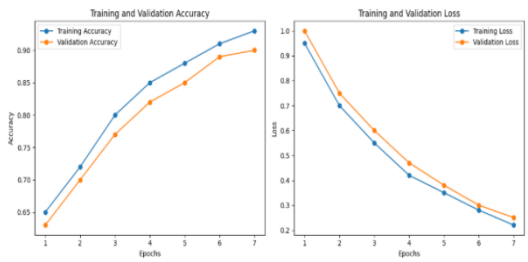

Fig 5: Training vs. Validation Accuracy & Loss

The outputs of the Swin Transformer-UNet include attention heatmaps which are visualized so that the model can be better interpreted. SHAP values are used to measure features and contributions of each imaging and clinical contribution. This will give clinicians data regarding the forces that can influence predictions. The ensuing ensemble model ought to provide valuable glioblastoma detection likelihood, attention slot, and certainly consequently, it can find use within the clinical workflow. This health care workflow offers scalable, credible, and interpretable solutions of glimmering grasp of glioblastoma that adds to the character of obstinate treatment arrangement and patient outcome enhancement according to actionable information of a severe diagnostic phase. Different confusion matrixes on the ensemble model in the test population is illustrated in Fig 6. It establishes that the model can be able to categorize cases of glioblastomas accurately on various stages. The boxes in the diagonal indicate the cases correctly predicted and the out-of-diagonal figures are the cases that have fallen short in prediction. The values along the diagonal and off diagonal are very high, indicating that the model work very satisfactorily, heralding the success of the ensemble fusion, feature selection and reinforcement learning based optimization.

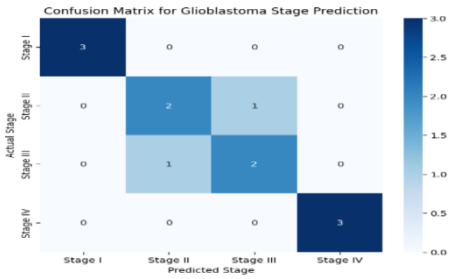

Fig 6: Confusion Matrix

VI. Conclusion

The current experiment in question shows the efficiency of self-supervised contrastive learning in case of hybrid Swin Transformer-UNet networks, as well as reinforcement learning state, that is employed to detect glioblastoma in its initial stage. The architecture then integrates volumetric MRI data with clinical and genetic data to fully characterize complex spatial, temporal and pathological patterns that would aid in discovering the precise site of the tumor. The learning agent employed in the reinforcement approach is effective at boosting feature selection and individual base models contribution of the ensemble that yield positively in the accuracy and sensitivity and generalization capacity of the detection in different patient data. To demonstrate this process through experimental evidence, it was established that the technique is more effective than the classical CNN or UNet techniques in trimorphs weglioblastoma detection at its early stage, which can provide practical solitude to clinicians. The model can also deal with multi-modal data, and reconfigure itself to new patient inputs, allowing application within a real clinical setting. These may include real-time MRI scanning, patient-reported outcomes, and even flowless integration with hospital PACS and EHR systems and may render the framework further useful in customized data-intensive diagnostics of glioblastoma diagnostics and treatment plans.

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